ASSIGNMENT-REGRESSION ALGORITHM

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same. As a data scientist, you must develop a model which will predict the insurance charges.

1. **Identify your problem statement**
   Want to predict the insurance charges.
   Stage 1: Machine Learning
   Stage 2: Supervised
   Stage 3: Regression
2. **Tell basic info about the dataset(Total number of rows and columns)**

   No. of rows = 1338
   No. of columns = 6
   No. of input features = 5
   Target variable = Charges

Column Name	Data type
Age	Integer
Sex	Categorical
BMI	Float
Children	Integer
Smoker	Categorical
Charges	Float

3. **Mention the pre-processing method if you're doing any (like converting string to number – nominal data)**
   Converting Categorical data into numerical data – Nominal (One hot encoding)
   Sex:
      Female -0
      Male- 1
   Smoker:
      Yes - 1
      No – 0

4. **Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.**
   I am using machine learning algorithm such as,
   1. Multiple Linear regression
   2. Support vector model
   3. Decision Tree
   4. Random Forest

5. **All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)**

1. **MULTIPLE LINEAR REGRESSION**

R2_score value = 0.7894790349867009

2. SVM-SUPPORT VECTOR MACHINE

S.NO	HYPER TUNING	LINEAR	RBF	POLY	SIGMOID
1.	-	-0.01010	-0.08338	-0.07569	-0.07542
2.	C=10	0.46246	-0.03227	0.03871	0.03930
3.	C=100	0.62887	0.32003	0.61795	0.52761
4.	C=1000	0.76493	0.81020	0.85664	0.28747
5.	C=2000	0.74404	0.85477	0.86055	-0.59395
6.	C=3000	0.74142	0.86633	0.85989	-2.12441

R2_score value = 0.86633

3. **DECISION TREE**

S.NO	CRITERION	SPLITTER	MAX_FEATURES	R_VALUE
1.	*squared_error*	*best*	*sqrt*	0.71410
2.	*squared_error*	*random*	*sqrt*	0.65011
3.	*squared_error*	*best*	*log2*	0.70268
4.	*squared_error*	*random*	*log2*	0.57289
5.	*squared_error*	*best*	*None*	0.70166
6.	*squared_error*	*random*	*None*	0.70615
7.	*absolute_error*	*best*	*sqrt*	0.72814
8.	*absolute_error*	*random*	*sqrt*	0.67105
9.	*absolute_error*	*best*	*log2*	0.73203
10.	*absolute_error*	*random*	*log2*	0.72852
11.	*absolute_error*	*best*	*None*	0.65901
12.	*absolute_error*	*random*	*None*	0.76278
13.	*poisson*	*best*	*sqrt*	0.67998
14.	*poisson*	*random*	*sqrt*	0.67743
15.	*poisson*	*best*	*log2*	0.78937
16.	*poisson*	*random*	*log2*	0.69373
17.	*poisson*	*best*	*None*	0.72852
18.	*poisson*	*random*	*None*	0.74211
19.	*Without parameter*			0.69287

R2_score value = 0.78937

4. RANDOM FOREST:

S.NO	N_ESTIMATORS	CRITERION	MAX_FEATURES	R_VALUE
1.	***10***	***squared_error***	***sqrt***	0.85200
2.	***100***	***squared_error***	***sqrt***	0.87102
3.	***10***	***squared_error***	***log2***	0.85200
4.	***100***	***squared_error***	***log2***	0.87102
5.	***10***	***squared_error***	***None***	0.83303
6.	***100***	***squared_error***	***None***	0.85383
7.	***10***	***absolute_error***	***sqrt***	0.85742
8.	***100***	***absolute_error***	***sqrt***	0.87106
9.	***10***	***absolute_error***	***log2***	0.85742
10.	***100***	***absolute_error***	***log2***	0.87106
11.	***10***	***absolute_error***	***None***	0.83506
12.	***100***	***absolute_error***	***None***	0.85200
13.	***10***	***poisson***	***sqrt***	0.85449
14.	***100***	***poisson***	***sqrt***	0.86801
15.	***10***	***poisson***	***log2***	0.85449
16.	***100***	***poisson***	***log2***	0.86801
17.	***10***	***poisson***	***None***	0.83139
18.	***100***	***poisson***	***None***	0.85263
19.		***Without parameter***		0.85230

R2_score value = 0.87106

6. Mention your final model, justify why u have chosen the same.

I selected Random Forest Regression because it gave the highest R^2 score and the lowest prediction error compared to the other models. Therefore, **Random Forest Regression** was selected as the final model because it achieved the best overall performance and accurately predicted insurance charges.

The Highest R2_score value is 0.87106